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BiWM: Advancing Open-Source Interactive Video World Models with Bidirectional Autoregression

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Interactive video world models commonly convert bidirectional video generators into causal autoregressive systems through control fine-tuning, autoregressive training, causal initialization, and few-step distillation. This pipeline is costly, while frozen causal histories accumulate errors that degrade long-horizon fidelity and controllability. We present **BiWM**, the first open-source full-stack training framework for bidirectional autoregressive video world models. **BiWM** retains full attention within each generated chunk and requires only two stages: camera/action-control fine-tuning and few-step Distribution Matching Distillation (DMD). Both stages converge within a few hundred optimizer steps on 8×H200 GPUs. The framework supports Wan2.1-T2V-1.3B, Wan2.2-TI2V-5B, HunyuanVideo-1.5-TI2V-8B, and LTX-2.3-22B, together with real-world camera control. Supervised and forward-KL anchors mitigate DMD mode collapse and preserve scene dynamics. **BiWM** provides a compact, reproducible path from pretrained bidirectional video models to interactive, controllable, and efficient world models.

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Code: https://github.com/LynnReal-AI/BiWM

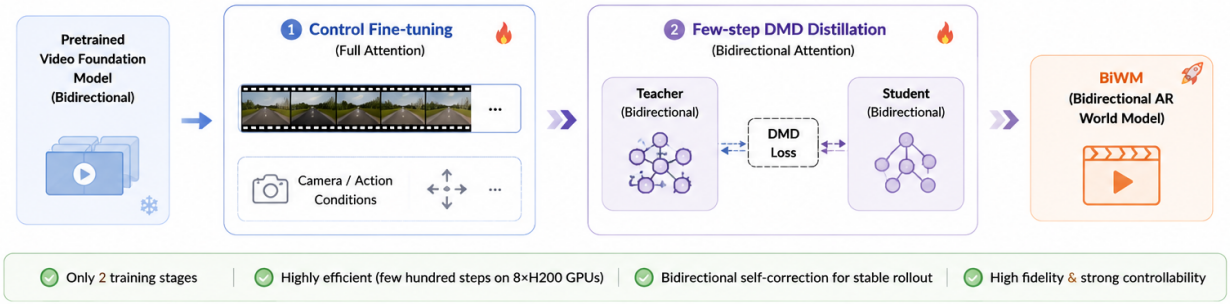


Figure 1 Overview of BiWM. From a pretrained bidirectional video foundation model, **BiWM** runs just two short training stages—camera/action control fine-tuning and few-step DMD distillation, both keeping full bidirectional attention—to obtain a bidirectional autoregressive interactive world model. The recipe uses only two training stages, is highly efficient (a few hundred steps on 8×H200 GPUs), self-corrects through bidirectional rollout for stable long-horizon generation, and attains high fidelity with strong controllability.

1 Introduction

Given a text or image prompt, contemporary video-diffusion models synthesize seconds to minutes of high-fidelity, temporally coherent footage (Brooks et al., 2024; Bao et al., 2024; Yang et al., 2025; Wan et al., 2025; Kong et al., 2024a; HaCohen et al., 2024). Re-purposing such a generator into an interactive video world model, one that continues a virtual world frame by frame under a stream of user actions (above all camera movements) and lets a person steer through it, has become a central goal of generative world modeling (Ball



Figure 2 Interactive world exploration with BiWM. Driven by discrete keyboard+mouse actions, BiWM lets a user explore a generated world. Text-to-video rollouts on Sekai-domain street scenes, each row navigated under a different constant discrete camera action—from top: *backward-right + yaw-right*, *right + yaw-left*, *forward-right + pitch-up, yaw-left*, and *forward, static look*. The bottom-left joystick overlay shows the action; the camera obeys each prescribed translation and look direction while preserving scene fidelity.

et al., 2025; Bruce et al., 2024; Sun et al., 2025; Tang et al., 2025; Mao et al., 2025; Ye et al., 2025; He et al., 2025; Xiang et al., 2025). Turning such an offline bidirectional generator into an autoregressive one that emits frames on demand is what makes these systems interactive and real-time, and is therefore the central technical challenge in building them.

Existing autoregressive world models fall into two families, distinguished by how the frames within a window attend to one another. **Causal** models (Team et al., 2026; Hong et al., 2025; Nam et al., 2026; HunyuanWorld, 2025) impose a causal mask so that each frame sees only its past;[1] their appeal is efficiency, since the past can be stored as a key–value (KV) cache and reused to accelerate the rollout. This caching, however, conceals a structural weakness: once a frame is frozen into the KV cache its representation can never be revised, so any error in the generated history is permanent and compounds as the rollout lengthens, eventually corrupting both the imagery and the model's response to control. The drift is more damaging for video diffusion than for language. An autoregressive language model predicts over a discrete vocabulary and re-quantizes onto valid tokens at every step, which endows it with an innate ability to absorb and correct small mistakes; a diffusion-based video model instead fits a continuous distribution over pixels, where sub-token deviations are never snapped back and accumulate unchecked until the scene collapses. The two failures compound: errors in the imagery and drift in the camera response reinforce one another, so a causal rollout under camera control degrades faster than either alone (Fig. 3). **Bidirectional** models, namely the Yume series (Mao et al., 2026a, 2025) and Matrix-Game-3.0 (Wang et al., 2026b), instead let every frame in a window attend to every other, exactly as the pretrained backbone does, and this is precisely what counteracts the drift: because earlier history latents remain visible to, and are refreshed alongside, the frames currently being denoised, the model continually self-corrects its own past, trading a modest amount of caching efficiency

[1] In practice many "causal" world models adopt a local window: attention is bidirectional within a short window of frames and causal only across windows, a compromise that recovers some visual quality. For clarity of exposition we set this refinement aside and speak of fully causal rollout; it does not affect the argument, since a window's representation is still frozen once it leaves the cache.



Figure 3 Why fully causal camera control collapses. A causal autoregressive baseline (Self-Forcing-style, continuous-pose control) rolling out an image-to-video clip under a walking camera trajectory (joystick overlay, bottom-left shows the commanded action). Left to right, errors frozen into the KV cache and drift in the camera response compound, and the scene degrades from a clean street into a washed-out, structureless frame. **BiWM**'s chunk-wise bidirectional rollout (self-correcting history) and discrete text-camera control are designed to avoid both failure modes.

for substantially better fidelity and controllability over long horizons. What makes this trade-off practical is few-step distillation: once a window denoises in only a handful of steps, retaining full bidirectional attention within it incurs little additional cost, and error resilience, rather than latency, becomes the deciding factor.

These bidirectional systems confirm the benefit empirically, reporting sharper frames and more stable long-horizon exploration than their causal counterparts. What the community still lacks, however, is an open, end-to-end recipe for the paradigm. minWM (Zhao et al., 2026a) has open-sourced a full-stack framework for causal interactive world models, yet no full-stack, open-source counterpart exists for the bidirectional autoregressive paradigm, which leaves its strong empirical results difficult to reproduce or extend.

We close this gap with **BiWM**, to our knowledge the first full-stack, open-source framework for building interactive video world models under the bidirectional autoregressive paradigm, designed to balance generation quality against generation speed. **BiWM** keeps the backbone's native full attention within each short window of latent frames and pays the autoregressive cost only across windows, conditioning each window on the history of those before it. Starting from a pretrained video foundation model, the recipe needs only two stages: a first stage that injects camera control by fine-tuning, and a second stage that directly performs few-step self-rollout DMD distillation, building on Self-Forcing (Huang et al., 2026) but in the chunk-wise bidirectional rather than causal setting (Yin et al., 2024b,a), after which the backbone becomes a camera- and action-controllable interactive world model. To counter the mode-seeking tendency of distribution-matching distillation, which otherwise collapses scene dynamics, we augment the DMD objective with anti-degradation terms, including supervised and mass-covering forward-KL anchors that preserve detail and motion diversity.

Importantly, **BiWM** is built for the resource budgets of academic research. It uses only two training stages, compared with four in minWM, and is inexpensive to train: camera control and DMD distillation jointly converge within a few hundred optimizer steps on 8×H200 GPUs, so that a complete world model can be validated and iterated within hours rather than weeks. Because none of the recipe is tied to a particular backbone, we provide full-stack training across architectures and modalities, including Wan2.1-T2V-1.3B, Wan2.2-TI2V-5B (Wan et al., 2025), HunyuanVideo-1.5-TI2V-8B (Kong et al., 2024a; Esser et al., 2024), and LTX-2.3-22B (HaCohen et al., 2024). The same framework also supports secondary fine-tuning of existing bidirectional autoregressive models such as Yume-1.5 and Matrix-Game-3.0, adapting them to new data distributions at low cost, and it enables real-world camera control, a regime that proves nearly uncontrollable under minWM. Fig. 1 summarizes the two-stage recipe. We position **BiWM** not as a competitor to causal frameworks but as their bidirectional complement in the same open-source design space, trading a small amount of per-window latency for fidelity, controllability, and substantially shorter training.

Contributions.

- We introduce **BiWM**, the first full-stack, open-source framework for interactive video world models under the **bidirectional autoregressive** paradigm, with full bidirectional attention within a chunk and autoregression across chunks, positioned as the bidirectional complement to causal frameworks such as minWM.
- We design a compact two-stage recipe (camera-control fine-tuning followed by few-step DMD distillation, versus four stages in minWM) that is deliberately suited to academic budgets: both stages jointly converge within a few hundred optimizer steps on 8×H200 GPUs. To prevent the mode-seeking collapse of distribution-matching distillation, we add supervised and mass-covering forward-KL anchors that preserve visual detail and scene dynamics.

- We demonstrate generality and reproducibility: **BiWM** provides full-stack training across Wan2.1-T2V-1.3B, Wan2.2-TI2V-5B, HunyuanVideo-1.5-TI2V-8B, and LTX-2.3-22B, additionally supports secondary fine-tuning of bidirectional autoregressive models (Yume-1.5, Matrix-Game-3.0) to new data distributions, and enables real-world camera control that proves nearly uncontrollable under minWM. We release code, scripts, and checkpoints together with reproducible component studies.

2 Related Work

Video (and audio–video) diffusion backbones. Large-scale diffusion transformers have become the generative prior behind most recent video world models, producing high-fidelity, temporally coherent clips across three broad architectural families: cross-attention conditioned designs (Wan et al., 2025; Yang et al., 2025; Bao et al., 2024), MMDiT designs that jointly attend over text and video tokens (Esser et al., 2024; Kong et al., 2024a), and, increasingly, models that generate synchronized audio and video (HaCohen et al., 2024). A property they all share is full bidirectional spatiotemporal attention over the entire clip—the very source of their fidelity. **BiWM** takes such a model as its teacher and retains this bidirectional attention within each generated chunk, rather than re-training it to be strictly causal. Since our recipe alters only the conditioning and the rollout, leaving the backbone's attention untouched, the same framework transfers cleanly across all three families.

Causal interactive world models. Most real-time world models convert an offline generator into a controllable, causal, low-latency roll-out engine (Ball et al., 2025; Bruce et al., 2024; Sun et al., 2025; Tang et al., 2025; Mao et al., 2025; Ye et al., 2025; Xiang et al., 2025; He et al., 2025; Hong et al., 2025; Shin et al., 2025; Feng et al., 2025), typically following the block-causal AR template of CausVid (Yin et al., 2025) and Self-Forcing (Huang et al., 2026) and distilling it to a few steps. minWM (Zhao et al., 2026a) packages this conversion via Causal Forcing (Zhu et al., 2026b; Zhao et al., 2026b) with continuous PRoPE (Li et al., 2026a) camera control. **BiWM** differs at the level of paradigm, keeping windows bidirectional, and in its controls (discrete text-camera actions), its objective (multi-objective short distillation), and its breadth of backbones.

Bidirectional interactive world models. A complementary line keeps each window bidirectional. The Yume series (Yume-1.0 (Mao et al., 2025) and Yume-1.5 (Mao et al., 2026a)) and Matrix-Game-3.0 (Wang et al., 2026b) let every frame in a window attend to every other and report sharper, more stable long-horizon rollouts than causal models, confirming the paradigm's benefit empirically. These systems remain hard to reproduce, however: at the time of writing none has released its training dataset or a complete training pipeline, and Matrix-Game-3.0 in particular still depends on a many-stage training recipe. **BiWM** targets exactly this gap, offering a fully open, two-stage recipe for the bidirectional paradigm together with its data and training code.

Camera-signal injection. Existing ways to inject camera control into a video diffusion backbone fall into two families. Absolute injection adds the camera signal onto the per-frame hidden state, either as a global, low-frequency control applied to the latent before the DiT (He et al., 2024), or layer-by-layer into every block's hidden state (Team et al., 2026). Because the pose is encoded as an absolute per-frame signal, this couples the temporal dynamics of the camera trajectory with those of the video itself, which tends to amplify error accumulation over long rollouts. Relative injection instead alters the inter-frame attention so that the interaction between two frames accounts for their relative pose; representative methods include CaPE (Kong et al., 2024b), GTA (Miyato et al., 2024), and PRoPE (Li et al., 2026a) (also adopted in HunyuanWorld 1.5 (HunyuanWorld, 2025)), as well as UCPE (Zhang et al., 2026) (adopted in SANA-WM). These are more robust, but they still inject the camera signal as a residual branch alongside the original attention; even with zero initialization, the residual perturbs the pretrained attention and induces a transient drop in visual quality early in training. Both families typically require a relatively heavy camera encoder or per-layer learnable injection modules, converge slowly, and lean on large batch sizes for training stability. In contrast, **BiWM** casts camera control as a conditioning task and injects the signal through the text space directly into the video tokens (Sec. 3.3): it adds no new learnable parameters, converges within roughly a hundred steps, and leaves the base generator's visual quality intact.

Few-step distillation and long-horizon memory. Distribution matching distillation (DMD) (Wang et al., 2023; Luo et al., 2023; Yin et al., 2024b,a) and consistency distillation (Song et al., 2023) compress many-step samplers to a handful of steps; applied to AR video, DMD with self-rollout (Yin et al., 2025; Huang et al., 2026; Yin et al., 2025) is the standard route to real-time generation. As its reverse-KL objective is mode-seeking, we pair it with a supervised regression (SFT) term and mass-covering forward-KL anchors, which together stabilize convergence and preserve visual detail and motion diversity. For long-horizon rollout, prior work compresses the ever-growing history in different ways: PackForcing uses a learned fixed-size memory (Mao et al., 2026b), while FramePack applies recency-weighted, multi-scale context packing (Zhang and Agrawala, 2025). These approaches are complementary to our focus on bidirectional autoregressive training.

3 Method

BiWM converts a pretrained multi-step bidirectional video-diffusion model into a few-step, camera-controllable, chunk-wise autoregressive world model. We emphasize at the outset that the entire training pipeline comprises exactly two stages (Fig. 1): **Stage 1**, camera-text pretraining, and **Stage 2**, multi-objective few-step distillation. There is no separate data-curation, or post-alignment stage. The central design choice, illustrated in Fig. 1, is to factorize generation into chunks that are denoised with full bidirectional attention internally yet produced autoregressively, each conditioned on the history of preceding chunks and on a stream of discrete camera-action tokens. We first establish notation (Sec. 3.1) and the data preprocessing that recovers continuous 6-DoF poses (Sec. 3.2), then describe camera-text control (Sec. 3.3), the bidirectional autoregressive rollout with sliding-window history conditioning (Sec. 3.4), the multi-objective distillation (Sec. 3.5), how a single recipe spans cross-attention, MMDiT, and audio–video backbones (Sec. 3.6), and the training budget (Sec. 3.7).

3.1 Problem Formulation

Let $\mathbf{x} = (\mathbf{x}^1, \ldots, \mathbf{x}^T)$ be the latent frames of a video produced by the VAE encoder of a foundation backbone, and let c be a (static-only) scene caption. We partition the T latent frames into B contiguous chunks of K frames each, $\mathbf{x} = (\mathbf{c}_1, \ldots, \mathbf{c}_B)$ with $\mathbf{c}_b = \mathbf{x}^{(b-1)K+1:bK}$. Autoregression simply means generating these chunks one after another: both the causal and the bidirectional paradigm share the same chunk-wise factorization

$$p(\mathbf{x} \mid c, \mathbf{a}) \;=\; \prod_{b=1}^{B} p\big(\mathbf{c}_b \;\big|\; \mathbf{c}_{<b}, c, \mathbf{a}_b\big), \tag{1}$$

where $\mathbf{a} = (\mathbf{a}_1, \ldots, \mathbf{a}_B)$ is a stream of discrete camera actions and $\mathbf{a}_b$ is the per-frame action sequence governing chunk b. What separates the two paradigms is not the chunk size but how each factor treats the history $\mathbf{c}_{<b}$ inside the attention.

Causal vs. bidirectional autoregression. A causal autoregressive model imposes a causal attention mask: while denoising chunk $\mathbf{c}_b$, the history is read from a frozen key–value cache, and the current chunk may attend to the past but the past may never attend to the present. Consequently the representation (the "state") of each history frame is fixed the moment it is produced and can never be revised. **BiWM** is instead bidirectional autoregressive: at each step it attends jointly and bidirectionally over the current chunk and its history, so the state of the history is itself conditioned on—and refreshed by—the chunk being generated. Because every already-generated frame remains free to update under the influence of the frames that follow it, the model continually re-interprets and self-corrects its own past, which is exactly what suppresses the error accumulation and camera drift of strict causality (Fig. 3). In short, what makes a model causal or bidirectional is simply whether its already-generated frames are still allowed to change—not how many frames it produces at a time. **BiWM** generates a short chunk of frames at each step and lets that chunk, together with the visible history, attend back and forth freely; the history is therefore re-encoded at every step and keeps being refined as generation moves forward. This is modestly more costly than caching the frozen history, but it is what lets the model stay sharp and on-trajectory while the rollout continues for arbitrarily long.

3.2 Data Preprocessing: Camera Trajectories to Discrete Actions

BiWM is grounded in continuous camera geometry: the discrete control of Sec. 3.3 is a quantization of true 6-DoF poses, not a hand-assigned categorical label. We draw on two complementary sources, each providing per-frame continuous camera trajectories that are later quantized into the action vocabulary.

Prescribed trajectories (OpenVid + WorldPlay). We directly reuse the open-source prescribed-trajectory data released by minWM (Zhao et al., 2026a), which samples still images from OpenVid (Nan et al., 2025) and uses WorldPlay (Sun et al., 2025) to generate videos that follow specified camera trajectories. Because the trajectory is prescribed rather than estimated, these clips carry exact ground-truth 6-DoF poses by construction, providing clean and diverse camera supervision at scale.

Real footage (Sekai). For real-world coverage we use the Sekai walking dataset (Li et al., 2026b) as our real-footage split: in-the-wild egocentric footage whose camera trajectory is not given and must be recovered. We follow the camera-annotation pipeline of SANA-WM (Zhu et al., 2026a), running a SLAM-style video pose engine (Huang et al., 2025) grounded with learned monocular geometry—a temporally consistent multi-view estimator (Wang et al., 2025) for structure together with a metric monocular model (Wang et al., 2026a) for absolute scale—and refining per-frame intrinsics through bundle adjustment. This recovers metric-scale per-frame camera-to-world extrinsics $T_i^{cw} \in SE(3)$ and intrinsics K_i for every real clip.

Filtering and captioning. Clips pass generic visual filters (aesthetic quality, motion magnitude, optical-flow consistency, scene-cut removal) and camera-specific filters on field of view, focal-length consistency, trajectory smoothness, and scale stability, which discard clips whose camera geometry is unreliable. Captions are written under a strict static-only instruction that describes objects, layout, and appearance but never camera motion, so that textual supervision cannot leak the trajectory and all motion is learned through the action stream.

From continuous poses to discrete actions. Finally, the recovered continuous trajectory is converted into the per-frame relative pose $(\Delta\mathbf{t}_i, \Delta\mathbf{R}_i)$ used by the quantizer of Sec. 3.3 (Eq. 2). We stress the relationship: **BiWM** remains faithful to continuous 6-DoF geometry throughout annotation, and discreteness enters only at the final quantization step, which maps the continuous motion onto the compact 81-class vocabulary so that it can be expressed as injectable text. The discrete vocabulary is thus a low-bandwidth, text-friendly encoding of real camera geometry, not a replacement for it.

3.3 Text-based Camera Control

The defining choice of **BiWM** is to treat camera control as a pure conditioning task carried entirely in the text space, adding no camera encoder and no new learnable parameters. Following the discrete camera-action encoding used in Yume-1.5 (Mao et al., 2026a), we quantize continuous 6-DoF camera motion into a translation×rotation vocabulary. Related systems such as HunyuanWorld-1.5 (HunyuanWorld, 2025) inject the resulting action into the diffusion time embedding. We instead express each action as ordinary text and inject it through the backbone's existing text-conditioning path, which leaves the pretrained input distribution intact and makes fine-tuning substantially more stable and data-efficient. The mechanism has four parts: quantization of camera motion into a discrete action vocabulary, one-time pre-encoding of that vocabulary, per-frame assembly of a camera-text + caption condition, and per-frame injection through the backbone's existing cross-attention. Let $\tau(\cdot)$ denote the (frozen) text encoder with output dimension d_t, and let $\mathbf{W}: \mathbb{R}^{d_t} \to \mathbb{R}^d$ be the backbone's existing text-projection (the same one applied to captions).

(i) Quantization of camera motion. We map continuous camera motion to a discrete action per latent frame. From the continuous trajectory recovered in Sec. 3.2, a frame's relative camera pose, a translation $\Delta\mathbf{t}_i$ and rotation $\Delta\mathbf{R}_i$ w.r.t. the previous keyframe, is fed to a direction-angle classifier, which assigns a translation class $g_{\mathrm{t}}(\Delta\mathbf{t}_i) \in \{0, \ldots, 8\}$ and a rotation class $g_{\mathrm{r}}(\Delta\mathbf{R}_i) \in \{0, \ldots, 8\}$ (class 0 = static when the magnitude is below an adaptive threshold; otherwise the nearest of eight canonical directions), combined into a single label

$$a_i \;=\; \underbrace{g_{\mathrm{t}}(\Delta\mathbf{t}_i)}_{\text{translation} \in \{0,\ldots,8\}} \times 9 \;+\; \underbrace{g_{\mathrm{r}}(\Delta\mathbf{R}_i)}_{\text{rotation} \in \{0,\ldots,8\}} \;\in\; \{0, \ldots, 80\}, \tag{2}$$

i.e. a 9-way translation (static / forward / backward / left / right / four diagonals) crossed with a 9-way rotation (static / pitch ± / yaw ± / four diagonals); see Fig. 4. The direction–magnitude decoupling makes the quantization robust to pose noise. Equivalently, keyboard/mouse logs or a textual pose string are parsed directly to the same labels, yielding a per-clip action stream $\mathbf{a} = (a_1, \ldots, a_{T_a})$.

(ii) One-time pre-encoding (separate encoding). Each of the 81 classes is tied to a fixed natural-language camera-motion phrase $\phi(a)$ (e.g. "*Camera moves forward. Camera yaws right.*"). Crucially, the camera vocabulary and the scene caption are encoded separately, and the vocabulary is encoded once at initialization rather than every step:

$$\mathbf{E}[a] \;=\; \tau\big(\phi(a)\big) \in \mathbb{R}^{L_a \times d_t}, \quad a \in \{0, \dots, 80\}, \qquad \widehat{\mathbf{E}} \in \mathbb{R}^{81 \times S_a \times d_t} \text{ (zero-padded to } S_a = \max_a L_a), \tag{3}$$

and stored as a frozen buffer. At run time we only gather from $\widehat{\mathbf{E}}$; the text encoder is never invoked on camera phrases during training, which removes them from the per-step training cost.

(iii) Per-frame condition assembly (text-feature concatenation). Let $\mathbf{C} = \tau(c) \in \mathbb{R}^{S_c \times d_t}$ be the caption embedding. For each latent frame i we concatenate its gathered camera-text with the caption and project through the shared text head,

$$\mathbf{Z}_i \;=\; \Pi_S\big(\big[\,\widehat{\mathbf{E}}[a_i]\;;\;\mathbf{C}\,\big]\big) \in \mathbb{R}^{S \times d_t}, \qquad \mathbf{H}_i \;=\; \mathbf{W}\,\mathbf{Z}_i \in \mathbb{R}^{S \times d}, \tag{4}$$

where $[\,\cdot\,;\,\cdot\,]$ is row-wise concatenation and Π_S truncates/zero-pads to the fixed text length S (we use S=512). Because the cross-attention applies no positional encoding to the context and no key masking, the result is invariant to the order of the camera and caption rows, which permits implementing Eq. 4 as a single padded, vectorized gather (camera-to-latent length is aligned by nearest-neighbour interpolation, $\mathbf{a} \leftarrow \mathrm{NN}(\mathbf{a}, T)$, which preserves discrete class boundaries that linear interpolation would blur).

(iv) Per-frame injection (decoupled streams). At every block, the camera-text enters through the backbone's existing cross-attention, with no new module. Each latent frame being generated is reshaped per frame and attends to its own condition: denoting frame i's N_p patch tokens by $\mathbf{X}_i$,

$$\mathbf{X}_i \;\mathrel{+}=\; \mathrm{CrossAttn}\big(\underbrace{\mathbf{X}_i}_{\text{query}},\; \underbrace{\mathbf{H}_i}_{\text{key/value}}\big), \tag{5}$$

so each frame attends to its own camera-text + caption $\mathbf{H}_i$. In Stage 1 the whole clip is denoised jointly and there is no history, so this per-frame injection is the only conditioning path: every latent frame receives its prescribed camera action through Eq. 5.

During autoregressive rollout (Stage 2, Sec. 3.4), the already-generated frames are summarized into history/memory tokens $\mathbf{X}^{\mathrm{mem}}$ (produced by one of the three history modes below) that condition the next chunk. These memory tokens carry no camera-text and attend to the caption only,

$$\mathbf{X}^{\mathrm{mem}} \;\mathrel{+}=\; \mathrm{CrossAttn}\big(\mathbf{X}^{\mathrm{mem}},\; \mathbf{W}\,[\mathbf{C}; \mathbf{0}]\big), \tag{6}$$

so the camera action is applied only to the chunk currently being generated, never re-applied to history. Keeping the two streams separate disentangles "what the world looks like" (caption) from "how the camera moves" (action), and prevents the history from leaking spurious camera cues into the future.

Parameter efficiency and training stability. Every operation above reuses pretrained components: the gather from $\widehat{\mathbf{E}}$ is parameter-free, and $\mathbf{W}$ and the cross-attention are the backbone's own. Unlike absolute or residual-relative injection, **BiWM** adds no parameters and no residual branch onto the self-attention, so at step 0 the model is exactly the pretrained generator conditioned on richer text. This is why control emerges in ~ 100 steps (Sec. 3.7) without the early-training quality dip or the large-batch requirement that prior residual camera-injection methods rely on.

3.4 Chunk-wise Autoregressive Rollout and Sliding-Window History Conditioning

To realize Eq. 1, chunk b is generated by the backbone conditioned on a memory representation of the already-generated history $\mathbf{c}_{<b}$. The already-generated clean latents are placed at noise level σ=0 in the sequence prefix, and the new chunk denoises conditioned on them through the backbone's native image-to-video timestep separation. To bound the cost, conditioning is restricted to a sliding window of the most recent clean latents together with a first-frame sink that anchors global layout; history beyond the window is discarded rather than compressed. The scheme is exact within the window and parameter-free, which makes it suitable for short to medium rollouts. The prefix retains the backbone's native patch-grid coordinates and true temporal indices for RoPE, avoiding spatial jumps or temporal-order changes at chunk boundaries. Because no compressed long-range memory is retained, distant context is lost once the rollout outgrows the window.

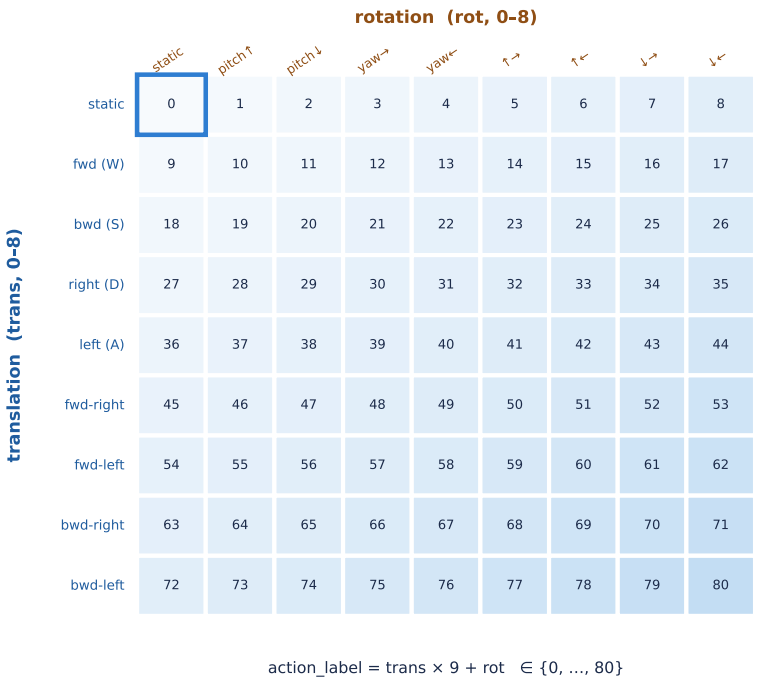


Figure 4 The 81-class discrete camera vocabulary as a 9×9 grid of translation $\times$ rotation; each cell is one action label (Eq. 2) and maps to a fixed camera-text phrase $\phi(a)$.

3.5 Multi-Objective Few-Step Distillation

The camera-text-pretrained model of Sec. 3.3 is a 50-step bidirectional sampler. Stage 2 distills it into a 4-step chunk-wise generator with self-rollout distribution-matching distillation. The training loop directly builds on Self-Forcing (Huang et al., 2026)—the generator rolls out its own chunks during training and is supervised by a DMD objective, closing the train–test gap of autoregressive video diffusion—and our key change is the paradigm: where Self-Forcing rolls out under a strictly causal mask, **BiWM** rolls out chunk-wise bidirectionally (full attention within each chunk, autoregression across chunks, Eq. 1). We use three copies initialized from the Stage-1 weights: a frozen real score s_{real} (the bidirectional teacher, evaluated with classifier-free guidance), an online fake score s_{fake} (a critic tracking the student's distribution), and the generator G_θ with velocity field v_θ, which self-rolls out a sequence $\tilde{\mathbf{x}}$ chunk by chunk. We adopt a flow-matching parameterization: for a clean latent $\mathbf{x}_0$, noise level $\sigma \in (0,1)$ and $\boldsymbol{\epsilon} \sim \mathcal{N}(0, I)$, the noised sample is $\mathbf{x}_\sigma = (1-\sigma)\mathbf{x}_0 + \sigma\boldsymbol{\epsilon}$, the velocity target is $(\boldsymbol{\epsilon} - \mathbf{x}_0)$, and the model's clean estimate is $\hat{\mathbf{x}}_0 = \mathbf{x}_\sigma - \sigma\, v_\theta(\mathbf{x}_\sigma, \sigma, c, \mathbf{a})$. The Stage-2 objective is a primary distribution-matching term regularized by two families of complementary anchors.

Primary: distribution-matching distillation. The leading term aligns the student to the teacher distribution through the asymmetric DMD gradient (Yin et al., 2024b, 2025; Huang et al., 2026)

$$\nabla_\theta\, \mathbb{E}_t\big[\mathrm{KL}\big(p_{\theta,t}(\tilde{\mathbf{x}}_t)\,\|\,p_{\text{data},t}(\tilde{\mathbf{x}}_t)\big)\big] = -\,\mathbb{E}_{\tilde{\mathbf{x}},\,t,\,\tilde{\mathbf{x}}_t}\Big[\big(s_{\text{real}}(\tilde{\mathbf{x}}_t, t) - s_{\text{fake}}(\tilde{\mathbf{x}}_t, t)\big)\,\frac{\partial \tilde{\mathbf{x}}}{\partial \theta}\Big], \tag{7}$$

where $\tilde{\mathbf{x}}_t$ is the noised student sample at level t, and $s_{\text{real}}, s_{\text{fake}}$ receive the same caption and camera-text conditions so controllability survives distillation. The critic s_{fake} is itself trained online (at an N:1 ratio against the generator) with a flow-matching velocity loss on the detached generator outputs. To keep self-rollout tractable, we retain the gradient on a single randomly chosen denoising step per chunk (Huang et al., 2026) and detach history across chunks, so the graph never spans the full rollout; a dynamic chunk-count curriculum concentrates compute on longer histories. Crucially, Eq. 7 is a reverse-KL objective and is therefore mode-seeking: minimized in isolation it tends to drop modes, manifesting as motion that decays toward a static scene or as high-frequency collapse. The two anchor families below counteract these failure modes.

Supervised anchor (SFT, low-σ velocity MLE). On the full real video latent $\mathbf{x}_0$ (all T frames, decoupled

from the per-iteration rollout length) we add a flow-matching velocity regression at low noise levels:

$$\mathcal{L}_{\mathrm{SFT}}(\theta) = \mathbb{E}_{\mathbf{x}_0,\,\sigma\sim\mathcal{U}(\sigma_{\min},\,\sigma_{\mathrm{sft}}),\,\boldsymbol{\epsilon}}\big[\,\big\|\, v_\theta(\mathbf{x}_\sigma,\sigma,c,\mathbf{a}) - (\boldsymbol{\epsilon}-\mathbf{x}_0)\,\big\|^2\,\big], \qquad \sigma_{\mathrm{sft}}\ \text{small}. \tag{8}$$

This design is inspired by the regression regularizer in the original DMD formulation (Yin et al., 2024b), which aligns the distilled model with teacher samples to mitigate the mode-seeking behavior of distribution matching. We instead align the student directly with real videos. At low σ, this provides a strong maximum-likelihood anchor that both reduces mode dropping and improves visual fidelity. It also allows practitioners to use a curated set of high-quality videos as supervised anchors during DMD, steering the distilled model toward the desired data distribution. Because the loss is computed on the complete video rather than on the (possibly short) rollout, it preserves long-video and motion modeling even when warmup uses a single block.

Forward-KL anchors (mass-covering). Our construction is inspired by the teacher-sample alignment in the original DMD formulation (Yin et al., 2024b) and related findings in HiAR (Zou et al., 2026). Both motivate anchoring the student to samples from a coverage-oriented reference distribution rather than relying solely on the mode-seeking reverse-KL DMD gradient. Building on these ideas, we adapt the objective to bidirectional autoregressive video and make task-specific refinements to the anchor construction, noise ranges, and loss weights. We therefore add a forward-KL term $\mathrm{KL}(p_{\mathrm{data}}\,\|\,p_\theta)$, which is mass-covering and penalizes dropping data modes (the source of low-motion degeneration). Minimizing forward KL reduces to an $\mathbf{x}_0$-regression (maximum-likelihood) objective on samples from the covered distribution, which we instantiate two ways. (a) Real forward-KL noises the full real video at high σ and regresses the student's clean estimate back to it,

$$\mathcal{L}_{\mathrm{rFKL}}(\theta) = \mathbb{E}_{\mathbf{x}_0,\,\sigma\sim\mathcal{U}(\sigma_{\mathrm{lo}},\,\sigma_{\mathrm{hi}}),\,\boldsymbol{\epsilon}}\big[\,\big\|\,\hat{\mathbf{x}}_0(\mathbf{x}_\sigma,\sigma) - \mathbf{x}_0\,\big\|^2\,\big], \qquad \sigma_{\mathrm{lo}} > \sigma_{\mathrm{sft}}, \tag{9}$$

complementing SFT: high σ governs global layout and motion, low σ governs detail. (b) Teacher forward-KL is data-free: the frozen teacher (with CFG) is rolled out along a dense ODE trajectory $\sigma{:}1\to 0$; for sampled trajectory anchors $(\mathbf{x}_{\sigma_a},\mathbf{x}_{\sigma_b})$ we read off the teacher's clean target by linear extrapolation, $\mathbf{x}_0^{\mathrm{teach}} = \mathbf{x}_{\sigma_a} - \sigma_a(\mathbf{x}_{\sigma_b}-\mathbf{x}_{\sigma_a})/(\sigma_b-\sigma_a)$, and regress the student's estimate at the same point:

$$\mathcal{L}_{\mathrm{tFKL}}(\theta) = \mathbb{E}\big[\,\big\|\,\big(\mathbf{x}_{\sigma_a} - \sigma_a\, v_\theta(\mathbf{x}_{\sigma_a},\sigma_a,c,\mathbf{a})\big) - \mathbf{x}_0^{\mathrm{teach}}\,\big\|^2\,\big]. \tag{10}$$

This transfers the teacher's full, mass-covering distribution without any real data, countering mode shrink and preserving rich camera-driven motion.

Total objective. The generator minimizes

$$\mathcal{L} = \mathcal{L}_{\mathrm{DMD}} + \lambda_{\mathrm{SFT}}\mathcal{L}_{\mathrm{SFT}} + \lambda_{\mathrm{rFKL}}\mathcal{L}_{\mathrm{rFKL}} + \lambda_{\mathrm{tFKL}}\mathcal{L}_{\mathrm{tFKL}}, \tag{11}$$

where each auxiliary term is optional and toggled by a single flag, letting practitioners trade stability for speed. Conceptually, DMD matches the teacher, the forward-KL anchors restore coverage, and SFT anchors fine detail to real data. With this objective the generator produces each K-frame chunk in 4 denoising steps and rolls out to 60 s and beyond.

Real-time event editing. A capability that the bidirectional paradigm makes natural—and that, to our knowledge, no prior open framework releases—is *event editing*: injecting a textual *event* into the scene while it is being explored. Each chunk is conditioned jointly on the event text and the discrete camera action, and because the history is continually re-encoded (Sec. 3.1), a user can introduce an event for the upcoming chunks and the bidirectional self-correction weaves it into the ongoing world coherently and in real time, then move on to the next event seamlessly. Fig. 5 shows **BiWM** realizing fantastical, prompt-specified events—glowing talisman streetlamps, rune-covered mechanical ladybugs, crystals breaking through the soil, a self-driving floating wheelchair—inside real street scenes while the camera moves under the joystick overlay. **BiWM** exposes this as a first-class feature, and we release the event dataset and scripts alongside the framework.

Image-to-video capability. Image-to-video capability is not obtained for free from a text-to-video checkpoint. For a T2V backbone, Stage 1 must include image-conditioned training so that the model learns to use the input frame together with the camera-action stream. At inference, the user-provided image is encoded into the first clean latent frame and placed as the initial history, after which the model continues the sequence under camera control. Native TI2V backbones such as Wan2.2-TI2V-5B already provide an image-conditioning path; for these models, Stage 1 retains and adapts the pretrained capability rather than learning it from scratch, and Stage 2 preserves it during distillation. The framework also supports mixed-task Stage 1 training over text-to-video, image-to-video, and video-to-video data when a shared checkpoint across these modes is desired.

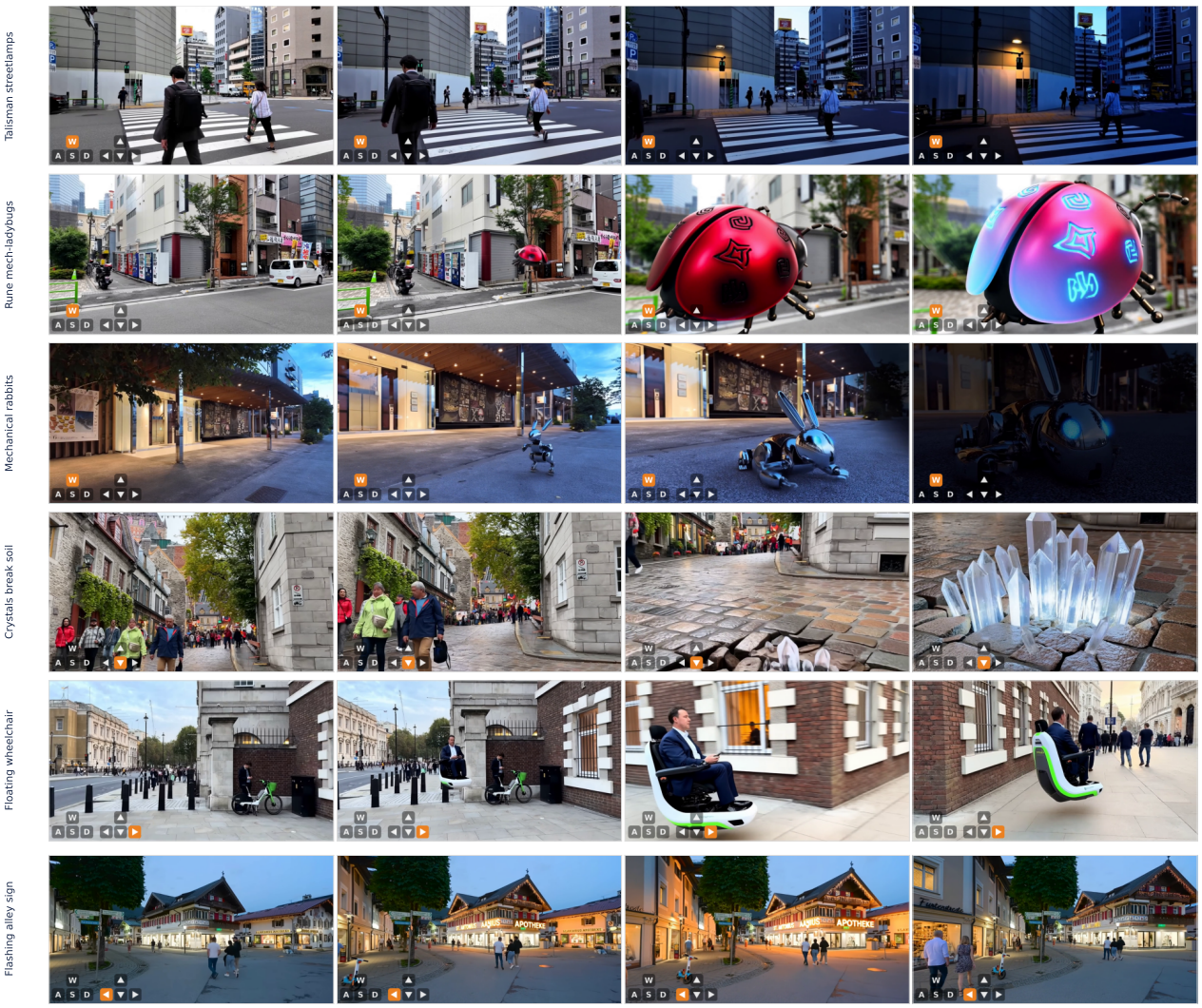


Figure 5 Event generation / real-time event editing. **BiWM** injects prompt-specified, fantastical events into real street scenes while the camera moves (joystick overlay, bottom-left). Top to bottom: glowing talisman streetlamps, rune-covered mechanical ladybugs that repel insects, mechanical rabbits, crystal clusters breaking through the soil, a self-driving floating wheelchair, and a flashing alley advertisement sign; each is shown over four frames. The event is specified purely by text and can be introduced or switched mid-rollout in real time. We release the event dataset and scripts. Qualitative illustration, not a benchmark.

3.6 Generality Across Architectures and Modalities

Nothing above is tied to a particular backbone: **BiWM** only requires a video-diffusion model whose attention can be evaluated chunk-wise and whose conditioning accepts per-frame text. We exploit this to instantiate the same two-stage recipe across three architecture families. On cross-attention backbones (Wan2.1-T2V-1.3B and Wan2.2-TI2V-5B (Wan et al., 2025)), camera-text enters through the existing text cross-attention. On an MMDiT backbone (HunyuanVideo-1.5 (Kong et al., 2024a; Esser et al., 2024)), where text and video tokens are jointly attended in double-stream blocks, the camera-text tokens are concatenated into the text stream and the chunk/history logic wraps the joint attention. On a joint audio–video backbone (LTX-2.3-22B (HaCohen et al., 2024)), the chunk groups paired audio and video latents so that each window denoises synchronized sound and vision together; the across-chunk history carries both streams, yielding an interactive world model that is audible as well as visible. Adapting to a new backbone amounts to providing an encoder for clean-latent history and a hook for per-frame camera-text, typically a thin adapter, while the camera vocabulary, rollout, and distillation code are shared.

3.7 Training Budget

A practical highlight of **BiWM** is how little training it needs. The chunk-wise bidirectional design keeps the backbone close to its pretrained prior, and the disentangled discrete camera-text is a low-dimensional signal to learn, so Stage 1 acquires reliable camera control in only $\sim$100 optimizer steps. Stage 2 distillation, with the SFT anchor of Eq. 8, converges in $\sim$200 steps. Both stages run on 8×H200 GPUs with gradient accumulation 4, so the entire pipeline completes in hours rather than days. Notably, there is no separate post-alignment stage; the two stages above constitute the whole recipe.

4 Implementation Details

We summarize the settings needed to reproduce **BiWM** beyond the recipe of Sec. 3. Real clips are truncated to 77 frames and captioned under a static-only instruction so that motion is carried solely by the discrete action stream; captions and the per-class camera-text bank are pre-encoded once for efficiency, and action-to-latent length alignment uses nearest-neighbour interpolation to preserve discrete class boundaries. In Stage 2 the generator self-rolls out chunk by chunk with one random denoising step per chunk retaining gradient (history detached across chunks), and a dynamic chunk-count curriculum concentrates compute on longer histories. Both stages run on 8×H200 GPUs with gradient accumulation 4, converging in $\sim$100 (Stage 1) and $\sim$200 (Stage 2) optimizer steps; each auxiliary loss is toggled by a single flag. For full reproducibility, all qualitative figures in this paper are generated from raw rollout frames by the released figure-generation script, and the released fine-tuning scripts reproduce both training stages and inference across the four backbones.

5 Results

Because **BiWM** is a framework rather than a single model, we characterize it through the function of each design choice rather than through a single benchmark number. We first record how the framework is instantiated and then analyze, component by component, the contribution and rationale of each element. A systematic quantitative study across backbones is ongoing and will accompany the code release; the goal here is to make the role of every component precise.

5.1 Instantiations

Backbones. The same recipe is instantiated on four backbones spanning three architecture families: cross-attention condition injection (Wan2.1-T2V-1.3B and Wan2.2-TI2V-5B (Wan et al., 2025)), an MMDiT design (HunyuanVideo-1.5 (Kong et al., 2024a)), and a joint audio–video design (LTX-2.3-22B (HaCohen et al., 2024)). Unless noted, clips are 77 frames (encoded to 20 latent frames) at each backbone's native resolution, grouped into chunks of K latent frames, with the distilled generator run at 4 denoising steps per chunk.

Data. Both stages share one format, a per-clip caption plus discrete camera actions: OpenVid+WorldPlay clips with prescribed-trajectory actions (Sec. 3.2), and the real-footage (Sekai) split, whose recovered poses are quantized into the 81 combined-action classes (9 translation × 9 rotation). Captions are produced by a vision-language model under a strict static-only instruction that describes scene appearance but never camera or object motion, so all motion supervision flows through the discrete action stream. These real pairs also supply the SFT and real forward-KL targets (Eqs. 8, 9).

Training budget. Both stages are short—$\sim$100 (Stage 1) and $\sim$200 (Stage 2) optimizer steps on 8×H200 GPUs with gradient accumulation 4, with no separate alignment stage; see Sec. 3.7 for why this short budget suffices.

5.2 Qualitative Results

These results illustrate **BiWM**'s qualitative behavior; they convey mechanism and visual quality rather than serving as quantitative benchmarks.

Long-horizon rollouts under camera control. Figure 6 shows chunk-wise text-to-video rollouts in which each window stays bidirectional and the history keeps updating as it is generated (Sec. 3.1, 3.4). Scene

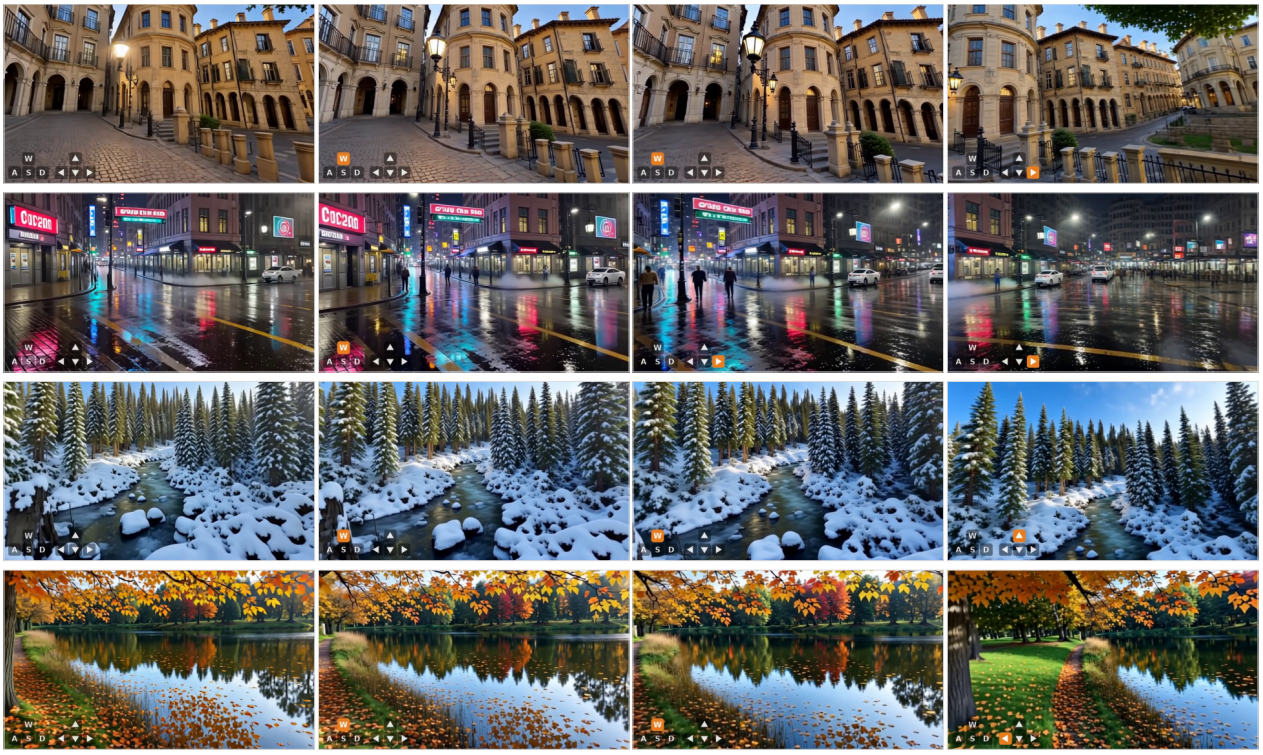


Figure 6 Illustrative T2V rollouts under camera control. Each row is a different text-prompted scene rolled out chunk-wise under its own camera motion (joystick overlay, bottom-left); keeping each window bidirectional preserves scene identity and geometry as the camera moves. Shown to illustrate the mechanism, not as a quantitative evaluation.

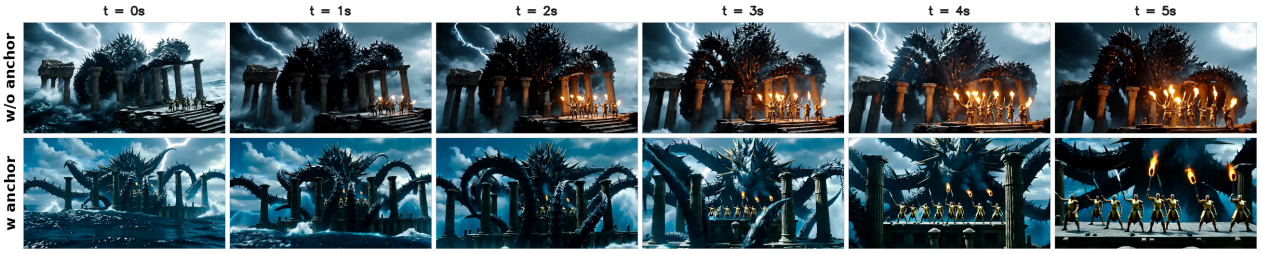


Figure 7 With vs. without the anchor losses. Same prompt, camera trajectory, and random seed; frames sampled every second from a 5 s rollout. The top row (w/o anchor loss) is the 4-step generator distilled with the DMD term only; the bottom row (w/ anchor loss) adds the SFT and forward-KL anchors. The anchor losses yield markedly sharper detail, higher contrast, and richer temporal dynamics, whereas DMD alone drifts toward a hazy, near-static rollout.

identity and geometry are preserved as the camera moves, and the visible history is re-encoded at every chunk.

Camera controllability. Figure 2 isolates the text-based control of Sec. 3.3: with a single constant discrete action per row, the camera obeys the prescribed translation and look direction while preserving scene fidelity.

Effect of the anchor losses. Figure 7 contrasts the generator distilled without the anchor losses (DMD term alone) against the one trained with them (adding the SFT and forward-KL anchors), under the same prompt, camera script, and random seed. Without the anchor losses the rollout is hazy and low in contrast, and its content barely changes over time, a direct symptom of the mode-seeking bias toward static, over-smoothed motion. With the anchor losses, high-frequency structure and contrast are restored and temporal dynamics increase markedly, with lighting and scene geometry evolving visibly across the horizon. The terms are complementary: the SFT anchor ties fine detail to real data, while the forward-KL anchors preserve motion coverage.

6 Conclusion

We presented **BiWM**, a recipe for bidirectional autoregressive video world models built on a chunk-wise factorization that retains the backbone's full bidirectional attention within each generated chunk while rolling out autoregressively across chunks. Two short training stages, camera-text pretraining and a multi-objective few-step distillation that augments distribution matching with auxiliary SFT and forward-KL terms, transform a 50-step bidirectional teacher into a 4-step chunk-wise generator steered by an 81-class discrete camera-action vocabulary. The recipe is economical: control emerges in $\sim$100 steps and distillation converges in $\sim$200 steps on 8$\times$H200 GPUs, with no separate alignment stage. It is also broad: a single recipe spans cross-attention (Wan2.1-1.3B, Wan2.2-5B), MMDiT (HunyuanVideo-1.5), and audio–video (LTX-2.3-22B) backbones, with the last yielding a world model that generates synchronized audio together with vision. For T2V backbones, image-to-video capability requires image-conditioned Stage 1 training, whereas native TI2V backbones such as Wan2.2-5B retain their pretrained image-conditioning path. We regard **BiWM** as the bidirectional point in the same design space as causal frameworks such as minWM, trading a small amount of per-window latency for fidelity, controllability, and markedly shorter training. Future directions include continuous and compositional action vocabularies, stronger long-horizon memory, richer audio–video control, and head-to-head benchmarking against causal recipes; we release code, checkpoints, and inference scripts to support them.

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